

Module 2 Lab Reflective Journal

Course: ITAI 1371

Assignment: Module 2 Lab Exercise

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Introduction

Lab 01 introduced me to the Iris dataset and showed me how Python can be used to explore and visualize data. This reflection covers what I learned, what I found challenging, and what I want to improve.

Description of Experience or Topic

The Iris dataset contains 150 samples, 4 features, and 3 classes: Setosa, Versicolor, and Virginica.

From the visualizations, I noticed that:

- Setosa generally has shorter sepals than the other two classes.
- Versicolor and Virginica generally have larger widths than Setosa.
- Virginica appears to be the largest overall in sepal length and width.

I would be interested in seeing whether these patterns stay the same with a larger sample size.

Personal Reflection

At first, the code felt overwhelming because there was a lot to understand. At the same time, much of the complexity was abstracted away by the Python libraries, so individual commands often did more than they appeared to. The visualizations helped me understand the dataset better because I could clearly see differences between the three flower classes. This showed me how useful visualization can be when analyzing data. I also gained more experience using Google Colab and Jupyter Notebook cells in VS Code. I am interested in seeing which environment I prefer as I continue working through the course.

Discussion of Improvements and Learning

This lab helped me become more comfortable with Python, notebook environments, and data visualization. In future labs, I want to spend more time experimenting with individual lines of code so I better understand what each function is doing. I can also apply what I learned by using visualizations to explore datasets before moving on to machine learning models.

Conclusion

Overall, this lab gave me a useful introduction to the Iris dataset and data visualization in Python. The code was overwhelming at first, but it became easier to understand as I worked through the lab and connected the code to the visual results.

Keira Burdine

Completing this lab helped me gain a better understanding of how data is organized, analyzed, and visualized when working with machine learning. Before beginning this assignment, I already had some experience using Python, NumPy, Pandas, and Matplotlib from previous assignments. I was familiar with concepts such as calculating the mean and median and creating basic graphs. However, one area that I initially struggled with during this lab was understanding the dataset itself. Seeing terms such as samples, features, classes, and dataset shape together was somewhat overwhelming at first. As I worked through each section of the lab, I began to understand what those terms represented and how they related to the information being analyzed.

Working with the Iris dataset helped make these concepts clearer to me. I learned that the dataset contained 150 flower samples, four different features, and three flower species. Understanding that the rows represented individual samples while the columns represented different measurements helped me see how a dataset is structured. Converting the information into a Pandas DataFrame also made the data easier for me to understand because I could see the information organized into a table rather than looking at the data as just numbers in Python.

One of the most useful parts of this lab for me was creating and interpreting the scatter plot. Instead of only looking at rows of measurements, I could visually see patterns in the data. The Setosa flowers appeared more separated from the other species, while Versicolor and Virginica had more overlap when comparing sepal length and sepal width. This helped me understand why data visualization is important. A graph can make relationships and patterns easier to recognize than simply reading raw values. It also made me wonder whether using petal length and petal width would make the differences between the species even easier to identify.

The statistical portions of the lab also helped strengthen concepts that I have previously learned. I used NumPy to calculate the mean and standard deviation of the sepal lengths and used Pandas to group the flowers by species and compare their average measurements. I was already familiar with calculating averages, but this assignment helped me understand how those calculations can be applied to an actual dataset. I also learned that standard deviation provides information about how much the values vary around the average. Seeing these concepts applied to real data made them more meaningful than simply calculating numbers by themselves.

Overall, this lab helped me become more comfortable working with datasets and understanding the purpose behind the different steps of data analysis. My biggest challenge was initially understanding what the dataset was showing me, but working through the lab step by step helped the information become much clearer. I now have a better understanding of how NumPy, Pandas, and Matplotlib can work together to analyze data, calculate statistics, and create visualizations. I also have a better

understanding of how these skills connect to machine learning because the computer needs organized and understandable data before it can recognize patterns or make predictions. I still have more to learn, but I feel more confident reading and interpreting a dataset than I did before completing this assignment.

Luis Adrian

Module 2 helped me understand how pandas, NumPy, and matplotlib can work together during the early stages of a machine learning project. This reflection also helped me identify the areas where I need more practice and what I can improve in future labs and projects.

One of the main concepts I learned in this lab was how pandas organize a dataset into a DataFrame. The DataFrame presented the information through rows and columns, making the data easier to read and analyze with Python. NumPy also helped me understand how numerical information can be summarized efficiently. For example, it allowed me to calculate the mean and standard deviation of the sepal length.

In the visualizations, I noticed several patterns while comparing the three flower species:

- Setosa generally had a wider sepal than versicolor and virginica.
- Versicolor and virginica had similar sepal lengths and widths, causing their points to overlap in the scatter plot.
- Setosa had the shortest sepal length of the three species.

These patterns were easier to recognize in the graphs than in the numerical table. This showed me that visualizations can make comparisons and relationships within a dataset easier to understand.

In this lab, I also learned that a notebook contains both code cells and Markdown cells. At first, I expected every cell to contain code, but I learned that Markdown is important for explaining and documenting an analysis. It helps me understand the purpose of the code and the meaning of its results.

Overall, this lab gave me an introduction to how datasets and DataFrames function in Python. I learned how a DataFrame can organize information into rows and columns that Python can process and analyze. I still need more practice with the different commands and functions, but this lab gave me a stronger foundation for future machine learning assignments and projects.

